Branch factor transported to weighted Proj

Formal proof packet

27 October 2026

This is a conditional polynomial and projective-stalk proof, not a target experiment or cost comparison. The editable diagram, gate CSV, Lean sources, and compiler logs are beside this PDF.

Checked statement

For polynomials f, u_1, u_2, n, U over a field and a rational point a, assume

$$f(a) = 0, \qquad n^2 - u_1 u_2 U = b^2 f, \qquad u_1(a)u_2(a)U(a) = 0.$$

The checked polynomial theorem specializes the norm identity at ordinate zero and uses the factor theorem to conclude

$$(X - a) \mid n.$$

The second checked theorem defines the coefficient map and x germ through the actual weighted-Proj branch-stalk isomorphism. It proves that the image of $X - a$ equals the projective $x - a$ germ and composes this with the norm theorem: the mapped polynomial $n(x)$ has a local factor $x - a$.

Norm identity plus rational branch-support root

$\Downarrow$ checked pointwise branch overlap: $n(a) = 0$

$\Downarrow$ checked polynomial factor theorem

$(X - a) \mid n$ and $n(x) = (x - a)t$ in the actual projective stalk

Verification and remaining gates

Direct Lean 4.34.1 compilation with Mathlib `d13f23b` exited zero for the polynomial source. Its theorem axiom audit lists only `propext`, `Classical.choice`, and `Quot.sound`; no `sorryAx`. Polynomial source SHA-256:

`7e583e29d15de7b4eba06d1cbb93a8eb52812c57039248791389d723317b8051.`

Projective source SHA-256: `20fce390a60291aab9950d24c99005ea8fc79d2b944a42ad91939bf40c932ce0.`

The concrete source-style norm theorem is separately checked. This packet does not identify the mapped polynomial with the compiled runtime numerator on every branch, prove that B is a local unit, or combine all local orders into the claimed global principal divisor and Pic^0 equation. The frozen 1,260-target rows and matched-rho ratios remain unchanged.